REBAL | Larger-data experiment

Tiny ImageNet classifier-rebalancing study
Experimental report - October 12, 2026

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Completed: 30 seed-by-imbalance cells, 120 trained linear classifiers, and 210 method evaluations across two fixed optimization budgets. Every evaluation uses the same 10,000 official validation images, held apart from classifier training and tuning.

Main finding

Feature SMOTE does not show a uniform advantage over simpler controls in this fixed-encoder setting. Its mean tail-F1 differences from random oversampling are +0.71, -0.17, and +0.00 percentage points at nominal imbalance ratios 10, 50, and 100. Against CE plus validation-selected logit adjustment, the corresponding differences are -0.99, -6.18, and -8.27 points.

Nominal ratio	SMOTE tail F1	ROS tail F1	CE + LA tail F1
10	33.90 +/- 2.31	33.19 +/- 2.22	34.89 +/- 2.27
50	20.63 +/- 2.21	20.80 +/- 1.99	26.81 +/- 2.39
100	13.92 +/- 2.53	13.92 +/- 2.34	22.19 +/- 2.32

Values are mean +/- sample standard deviation, in percent, over five paired seeds. These results use the primary 4,000-update budget. The report includes all methods and the prespecified shorter-budget sensitivity, without selecting a preferred budget after seeing evaluation scores.

What this result establishes

The experiment measures classifier weighting, oversampling, feature interpolation, and prior correction on shared frozen representations. Tiny ImageNet provides twice CIFAR-100's source training images and classes, and four times the pixels per image. The actual long-tailed training subsets contain 19,168 to 34,765 images; the study did not train a classifier on all 100,000 source images at once.

Scope limit: the encoder is an ImageNet-pretrained ResNet18. Tiny ImageNet is derived from ImageNet, so pretraining may overlap its images and classes. This is a controlled transfer study, not an end-to-end REBAL-F replication or clean from-scratch generalization test. It does not test the equalization regularizer, cGAN effects, or learned-representation improvement.

1. Data and experimental design

The official Stanford Tiny ImageNet archive supplies 100,000 labeled training images and 10,000 labeled validation images in 200 classes at 64 x 64 pixels [1]. The unlabeled official test partition is not evaluated. Official validation is treated as the final evaluation partition throughout.

Duplicate audit and sample allocation

Decoded RGB SHA-256 hashes identified seven training images identical to final-evaluation images and 20 additional duplicate training rows. All 27 were removed before sampling. The remaining 99,973 training images contain at least 495 unique examples per class. A seeded internal holdout reserves 50 images per class (10,000 total); the common maximum fitting count is therefore 445 per class. No identical decoded pixel array crosses fitting, internal holdout, or final evaluation after this exclusion.

Nominal ratio	Real fitting images	Min / max per class	Actual max/min	SMOTE or ROS pool
10	34,765	44 / 445	10.114	89,000
50	22,309	8 / 445	55.625	89,000
100	19,168	4 / 445	111.250	89,000

Counts follow floor(445 x rho^(-rank/199)). Each seed randomly permutes semantic class ranks; ranks and holdouts remain fixed across imbalance levels. The smaller fitting sets are nested inside the less imbalanced sets for that seed. Frequency regions follow the original scalar-metric convention: 66 head, 67 middle, and 67 tail classes by assigned rank. They are rank thirds, not universal many/few-shot count thresholds.

Matched controls

Method	Intervention on the shared frozen features
Natural CE	Linear classifier on the natural fitting distribution.
CE + logit adjustment	Same CE head; subtract tau log empirical training prior.
Class-balanced CE	Effective-number weighted CE, beta=0.999; mean-one class weights.
Random oversampling	Duplicate minority-class real features to 445 examples per class.
Feature SMOTE	Interpolate within-class neighbors to 445 per class; k=min(5,min count-1).
SMOTE + logit adjustment	Same SMOTE head; tune prior correction separately.
Nearest class mean	Normalize real features, average by class, normalize centroids; cosine decision.

The trainable heads use raw 512-dimensional features. NCM uses its explicitly defined cosine normalization, matching the repository's probe convention. No scaler, PCA, data cleaner, or feature extractor is fit to final-evaluation images.

2. Complete primary results

Final evaluation after 4,000 updates. Mean +/- sample SD over seeds 0-4; all values are percentages. The evaluation set is balanced (50 images per class), so top-1 accuracy equals macro recall. Zero-denominator class F1 is defined as zero.

Ratio	Method	Accuracy	Macro F1	Head F1	Middle F1	Tail F1
10	CE	35.98 +/- 0.41	35.23 +/- 0.42	38.00 +/- 2.10	36.33 +/- 1.39	31.39 +/- 2.00
10	CE + LA	37.77 +/- 0.21	37.48 +/- 0.28	40.84 +/- 2.07	36.77 +/- 1.40	34.89 +/- 2.27
10	CB	38.02 +/- 0.16	37.61 +/- 0.23	41.94 +/- 1.97	37.17 +/- 1.76	33.78 +/- 2.20
10	ROS	36.99 +/- 0.27	36.57 +/- 0.27	40.28 +/- 2.15	36.31 +/- 1.65	33.19 +/- 2.22
10	SMOTE	37.39 +/- 0.38	36.99 +/- 0.28	40.40 +/- 2.06	36.71 +/- 1.38	33.90 +/- 2.31
10	SMOTE + LA	37.43 +/- 0.34	37.39 +/- 0.22	41.10 +/- 2.27	36.83 +/- 1.52	34.31 +/- 2.45
10	NCM	33.63 +/- 0.14	32.73 +/- 0.12	34.23 +/- 2.56	32.40 +/- 0.93	31.58 +/- 2.26
50	CE	29.13 +/- 0.14	26.62 +/- 0.56	33.61 +/- 1.23	31.06 +/- 1.74	15.30 +/- 2.04
50	CE + LA	33.14 +/- 0.48	32.86 +/- 0.75	38.85 +/- 1.78	33.03 +/- 1.86	26.81 +/- 2.39
50	CB	33.60 +/- 0.24	31.74 +/- 0.37	39.01 +/- 1.43	33.67 +/- 1.54	22.65 +/- 1.81
50	ROS	31.57 +/- 0.21	29.70 +/- 0.39	36.45 +/- 1.72	31.93 +/- 1.72	20.80 +/- 1.99
50	SMOTE	31.76 +/- 0.36	29.86 +/- 0.53	36.46 +/- 1.51	32.57 +/- 1.71	20.63 +/- 2.21
50	SMOTE + LA	33.33 +/- 0.36	32.75 +/- 0.50	39.14 +/- 2.26	32.99 +/- 1.78	26.21 +/- 2.56
50	NCM	31.61 +/- 0.31	30.45 +/- 0.44	33.64 +/- 2.09	31.05 +/- 1.34	26.72 +/- 2.42
100	CE	26.27 +/- 0.56	22.86 +/- 0.37	31.91 +/- 1.54	28.06 +/- 1.93	8.75 +/- 1.67
100	CE + LA	30.67 +/- 0.28	30.14 +/- 0.28	37.54 +/- 2.14	30.80 +/- 1.69	22.19 +/- 2.32
100	CB	31.35 +/- 0.36	28.45 +/- 0.44	37.53 +/- 1.57	31.92 +/- 1.72	16.04 +/- 2.64
100	ROS	28.94 +/- 0.23	26.07 +/- 0.25	34.39 +/- 1.62	30.03 +/- 1.44	13.92 +/- 2.34
100	SMOTE	29.14 +/- 0.32	26.20 +/- 0.41	34.50 +/- 1.40	30.29 +/- 1.70	13.92 +/- 2.53
100	SMOTE + LA	31.15 +/- 0.34	30.23 +/- 0.34	38.23 +/- 1.98	31.05 +/- 1.77	21.53 +/- 2.48
100	NCM	30.02 +/- 0.27	28.41 +/- 0.37	32.93 +/- 2.09	30.16 +/- 1.51	22.20 +/- 2.32

Reading the table

A gain in tail F1 can trade off with head performance, and the best method by overall accuracy can differ from the best by tail F1. The random-oversampling comparison isolates interpolation from a change in class prior. CE + LA is a stronger comparison than natural CE alone because it corrects the learned prior without synthesizing training features.

SMOTE + LA is reported to inspect whether prior adjustment composes with interpolation. These rows are descriptive extensions; the prespecified six-comparison inference family concerns unadjusted SMOTE versus ROS and CE + LA on tail F1.

3. Imbalance and optimization budget

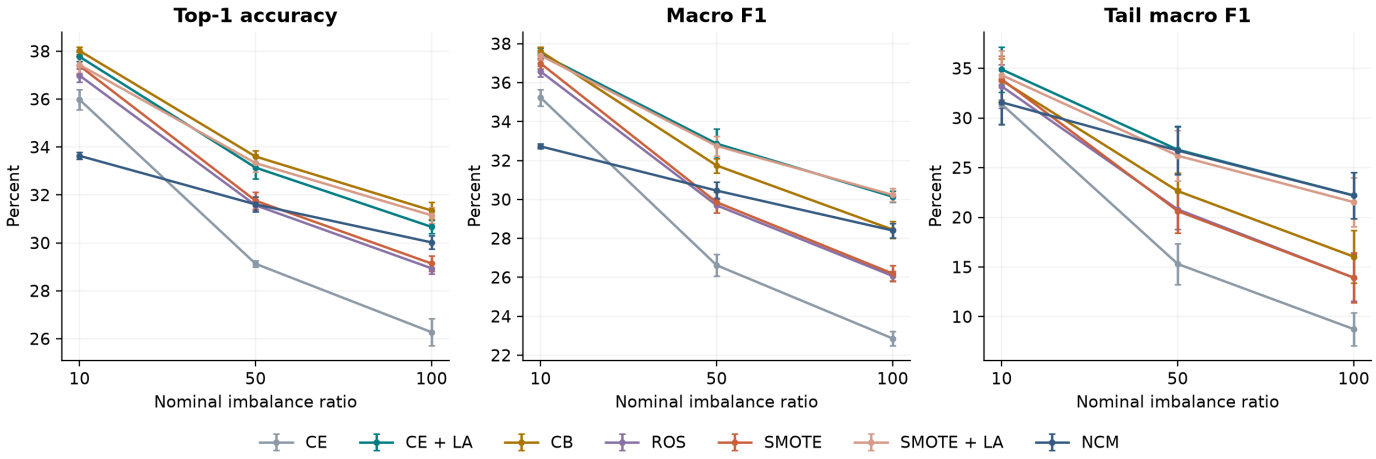


Figure 1. Performance versus nominal imbalance ratio. Error bars show sample SD across five seeds, not a confidence interval over new image populations. All methods share the same frozen encoder and evaluation images.

Prespecified 1,000-update sensitivity

Ratio	Method	Tail F1 at 1,000	Tail F1 at 4,000	Change (points)
10	CE + LA	35.69 +/- 2.64	34.89 +/- 2.27	-0.80
10	ROS	34.76 +/- 2.40	33.19 +/- 2.22	-1.57
10	SMOTE	35.41 +/- 2.42	33.90 +/- 2.31	-1.51
50	CE + LA	27.04 +/- 2.30	26.81 +/- 2.39	-0.23
50	ROS	23.65 +/- 2.55	20.80 +/- 1.99	-2.85
50	SMOTE	23.55 +/- 2.73	20.63 +/- 2.21	-2.92
100	CE + LA	21.47 +/- 2.89	22.19 +/- 2.32	+0.72
100	ROS	15.84 +/- 2.49	13.92 +/- 2.34	-1.91
100	SMOTE	15.93 +/- 2.52	13.92 +/- 2.53	-2.00

The shorter run was specified before the first classifier evaluation and uses the same splits, initializations, draw streams, and hyperparameters. It provides a training-budget check rather than an independent dataset replication. The complete seven-method sensitivity results are included in the machine-readable results. Neither budget is assumed to be optimal or fully converged.

More updates did not uniformly help. At nominal ratio 100, SMOTE tail F1 changes from 15.93% at 1,000 updates to 13.92% at 4,000. Most trained methods also have lower mean top-1 accuracy at the longer budget. This is evidence of sensitivity to optimization budget, not proof that either checkpoint is optimal.

4. Paired effects and uncertainty

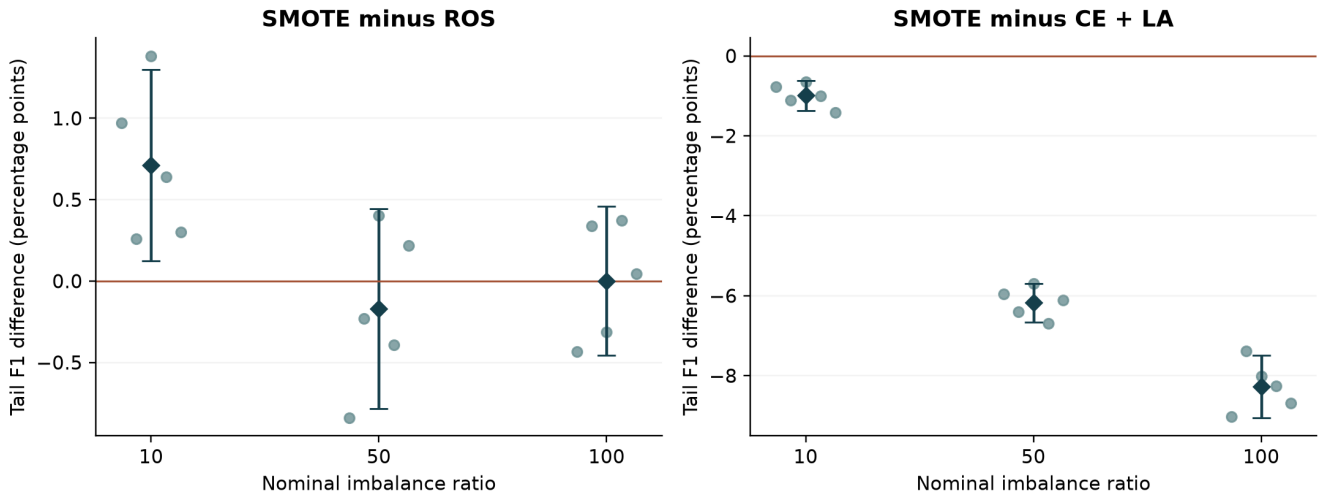


Figure 2. Dots are individual paired seed differences; diamonds and bars show their mean and unadjusted Student-t 95% interval. Positive values favor feature SMOTE.

Ratio	Comparison	Delta (points)	95% t interval	t p, Holm	Exact p, Holm
10	SMOTE - ROS	+0.71	[+0.12, +1.30]	0.085	0.375
10	SMOTE - CE + LA	-0.99	[-1.36, -0.62]	0.007	0.375
50	SMOTE - ROS	-0.17	[-0.78, +0.45]	0.976	1.000
50	SMOTE - CE + LA	-6.18	[-6.66, -5.70]	2.19e-05	0.375
100	SMOTE - ROS	+0.00	[-0.46, +0.46]	0.993	1.000
100	SMOTE - CE + LA	-8.27	[-9.06, -7.49]	4.05e-05	0.375

Five seeds are the paired units. Each seed changes the semantic frequency ordering, selected training images, and classifier initialization. Holm adjustment is applied to the six prespecified ratio-by-control comparisons, separately for the t-test and exact sign-flip test. The displayed confidence intervals are not multiplicity-adjusted.

With five paired seeds, the smallest possible two-sided exact sign-flip p-value is 0.0625 before adjustment. A small parametric t-test p-value therefore depends on assumptions that are difficult to assess with n=5. Interpret the observed effect sizes and direction across seeds as exploratory evidence; do not equate nonsignificance with equivalence. Shared pretraining and evaluation images are not additional independent replicates.

5. Source audit and reproducibility repairs

The repository was inspected at base commit `60e801a79e09a4911468af77d34ebf1726749a52`. Both main and kishor_dev pointed to that commit on retrieval. Existing aggregate CIFAR results were preserved. The original TensorFlow training experiments were not rerun, so the following findings describe the inspected source rather than proving historical numerical results wrong.

Bundled baseline inherits equalization

framework.py trains its baseline with uniform class weights but leaves the global equalization toggle active. Its usual bundled baseline is therefore not identical to standalone plain CE. This matters for claims about learned representation quality.

Preprocessing and development splits differ

The framework fits ECOD before carving out development data; standalone baselines omit ECOD and use a different splitting rule. Matching architecture and schedule alone does not establish matching effective data.

The feature-SMOTE ablation changes several factors

Turning feature SMOTE off invokes pixel SMOTE, changes the image distribution for backbone training, and removes cRT. It cannot isolate the effect of removing feature interpolation alone.

Generator and probe provenance is incomplete

The sample-bank preparation path does not reserve downstream development examples. Probe defaults target the main directory, whereas no-cGAN REBAL-F lives under ablate_cgan. Saved generator and checkpoint split identities are needed for strong reproduction claims.

Repairs implemented and tested

The experiment launcher now resolves the repository root correctly, isolates inherited protocol overrides, checks dataset/seed/method/ablation identity on resume, and returns failure status when a run fails. External sample-bank paths remain supported. Aggregators exclude incompatible records before producing corrected tables and keep legacy tables separate. The per-class CSV writer now uses the same region boundaries as scalar metrics for both CIFAR-10 and CIFAR-100.

Validation: 26 regression/protocol tests passed, including a miniature complete seven-method experiment, resume integrity, duplicate isolation, split nesting, aggregation filters, and CLI operation outside the repository. Independent verification recomputed the experimental metrics from saved predictions using scikit-learn and checked the split and optimization records. The attached source audit contains immutable source links and the remaining methodological issues.

6. Exact execution and reproduction

Frozen features were extracted with torchvision ResNet18 IMAGENET1K_V1 in evaluation mode on the Mac's MPS GPU. Inputs remained at native 64 x 64 resolution, converted to RGB, scaled by 1/255, and normalized with ImageNet mean/std. The model's classifier was replaced with identity. The usual 224-pixel pretrained inference crop was not used; that adaptation limits comparisons with published pretrained-model benchmarks.

All trainable heads use a new 512-to-200 linear layer with bias, identical seed-specific initialization, SGD learning rate 0.01, momentum 0.9, weight decay 0.0005, batch size 512, and uniform draws with replacement from the chosen pool. Learning rate is constant and the final fixed-update weights are evaluated. CB averages the mean-one class-weighted per-example losses, retaining the repository's weighting convention. Its effective gradient scale can differ from unweighted CE.

Heads ran deterministically on CPU with four threads; a synthetic timing check selected CPU because it was faster for this small linear model. The primary run gives 2,048,000 sample exposures per trained head; the shorter run gives 512,000. Equal update counts avoid giving enlarged SMOTE/ROS pools extra optimization merely because they contain more rows.

For CE + LA and SMOTE + LA, tau is chosen from {0, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2} by internal-holdout macro recall, with lower tau winning a tie. Priors use real fitting counts, not oversampled counts. There is no evaluation-based checkpoint or hyperparameter selection.

Commands, after downloading the official archive

```
python -m venv .venv-tiny
. .venv-tiny/bin/activate
pip install -r requirements-tiny.txt
python scripts/extract_tiny_features.py \
  --archive data/tiny-imagenet-200.zip \
  --out data/tiny_resnet18_64_features.npz --device mps
python scripts/benchmark_tiny_features.py \
  --cache data/tiny_resnet18_64_features.npz \
  --out experiments/tiny_primary_4000 --steps 4000 --device cpu
python scripts/benchmark_tiny_features.py \
  --cache data/tiny_resnet18_64_features.npz \
  --out experiments/tiny_sensitivity_1000 --steps 1000 --device cpu
python -m unittest discover -s tests -v
```

Provenance checksums

Official archive SHA-256

`6198c8ae015e2b3e007c7841da39ec069199b9aa3bfa943a462022fe5e43c821`

Frozen-feature cache SHA-256

`b180a355894bd50df295393f2c5877ff7d2b11fdb98528f444dd16a40100ef6e`

Benchmark source SHA-256

`315a900f5d4a4bb48db66e8508b3f19b902df0b906200c2091efada95d839309`

Recorded environment: Python 3.12.14; imbalanced-learn 0.14.2, numpy 2.5.4, scikit-learn 1.9.1, scipy 1.18.1, torch 2.14.1. Full commands, timestamps, hashes, splits, predictions, classifier parameters, per-class metrics, and tau curves are retained in the evidence package.

7. Interpretation, limits, and next experiment

The useful result is the controlled comparison: any apparent advantage of feature interpolation must survive a matched random-oversampling control and a validation-tuned prior-correction baseline. On these features, the observed benefit depends on the imbalance regime and optimization budget. The report presents the complete grid so that a favorable individual row cannot stand in for a general claim.

Pretraining and representation scope

The encoder may have seen Tiny ImageNet source images during ImageNet training. Exact duplicate removal protects the new classifier splits, but does not resolve pretraining overlap or near duplicates. No backbone is trained here; the study cannot confirm REBAL-F representation gains, equalization effects, or cGAN harm.

Additional calibration labels

The balanced internal holdout supplies 50 labels per class even when fitting has only four tail examples. This deliberate policy provides stable tuning coverage, but is more generous than a truly scarce-tail setting. All LA claims are conditional on that calibration budget; naturally imbalanced or reduced-label calibration is an important follow-up.

One dataset and five seed replications

Uncertainty is conditional on the chosen encoder, native-resolution transform, final evaluation set, and predefined optimization settings. It does not cover new datasets, independently trained encoders, unseen categories, or all plausible hyperparameters. The two budgets are not proof of convergence.

Next decisive experiment

Repair and freeze the full learning protocol, then train from scratch on duplicate-cleaned Tiny ImageNet with shared split manifests. Compare CE, class-balanced CE, class-balanced plus equalization, and matched ROS/SMOTE cRT. Run at least ten paired seeds with the same backbone, augmentation, update budget, and final-checkpoint rule; evaluate NCM and identical retrained heads on every backbone. A separate pretrained-transfer study on a dataset without known source overlap would test portability.

Primary references

1. Le and Yang (2015), Tiny ImageNet Visual Recognition Challenge
2. Chawla et al. (2002), SMOTE
3. Cui et al. (2019), Class-Balanced Loss
4. Kang et al. (2020), Decoupling Representation and Classifier
5. Menon et al. (2021), Long-tail Learning via Logit Adjustment
6. Torchvision ResNet18 model and weight documentation